DEPARTMENT OF
CONSUMER AFFAIRS

BIOPACKATHON 2026

Reinventing Sachets for a Litter-Free India

BIODEGRADABLE PACKAGING SOLUTION CHALLENGE FOR PAN MASALA & GUTKHA SACHETS

Solution Submission Template

Participant Details

Name: Mitti Se Pyaar (MittiSeal — From the waste of its own industry)
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Solution Overview

1. Title of the Packaging Solution: MittiSeal
2. Brief Description of the Packaging Solution:

India discards approximately 275,000 tonnes of areca husk every year from the same crop that supplies pan masala's primary ingredient. MittiSeal turns that waste into the packaging. MittiSeal is a 35 gsm paper sachet on a 30% areca-husk + 70% sugarcane-bagasse pulp blend (both agri-waste, both Indian). The base is layered with four natural-polymer coatings in a defined stack: a 2 gsm cellulose nanofibril (CNF) interlayer from ICAR-CIRCOT Mumbai (moisture/oxygen barrier), a 3 gsm food-contact chitosan inner coating from Mahtani Chitosan Veraval (oxygen barrier + antimicrobial), and a 4 gsm dewaxed orange shellac outer coating in aqueous-ammoniacal dispersion from Indian lac producers (India holds >85% of world shellac).

No synthetic polymer. No PLA. No metallized film. Critically, no separate adhesive layer. The hermetic lap seal is formed by brief infrared reactivation of the outer shellac itself, which autohesively fuses to itself under VFFS jaw compression — operationalizing Chauhan et al. (IIT Delhi, *IJBM* 2025, DOI 10.1016/j.ijbiomac.2025.143172) on Nichrome PV-215 multilane equipment at 300–600 sachets/minute. Every gram is a naturally-occurring polymer outside the PWM Rule 3(o) enumeration and generates no microplastic, complying with the 14 March 2024 PWM Amendment.

An off-line seed-paste zone on the outer face embeds native tulsi and marigold seeds in a starch binder. A littered sachet germinates into local flora instead of persisting as waste.

3. Status of Solution :

- Prototype-ready
- Already developed

4. Material Composition (including layers, coatings, adhesives) :

5. Packaging Design and Structure (drawings/photographs):

5. Target Sachet Volume / Size :

6. Cost per Unit/Sachet:

Challenge Requirement Mapping

1. Biodegradability

- Describe how the packaging biodegrades under ambient soil, freshwater, and marine environment *(approx. 250 words) (provide component wise biodegradability and environment conditions in which the packaging biodegrades along with timeframe for biodegradation and end products of biodegradation)* * :

  Note - indicate if *evidence on biodegradability of any component is not available.*

- Attach supporting documents *(data/test reports/certifications or any other evidence, as available)*

2. Food Safety & Compliance

- Describe how the packaging complies with Food Safety and Standards (Packaging) Regulations, 2018, as amended and Plastic Waste Management Rules, 2016, as amended *(approx. 250 words):*

- Attach supporting documents *(data/test reports/certifications or any other evidence, as available)*

3. Functional Performance

- Provide data about functional properties of the packaging solution :

Properties	Observed value
Mechanical strength	
Barrier properties	
Aroma retention	
Shelf-life stability	
Any other parameter	

- Attach supporting documents *(data/test reports/certifications or any other evidence, as available)*

4. Scalability & Economic Viability

- Describe the manufacturing process, mass production feasibility, and cost estimates *(approx. 250 words):*

- Attach supporting documents *(such as pilot-scale data, feasibility studies etc. as available)*

5. Limitations of the developed solution and future improvements required *(approx. 250 words):*

Additional Notes / Innovations

Include any other information, unique features, or innovations that strengthen the submission.

Declaration

I/We hereby declare that the solution submitted complies with the terms and conditions of BIOPACKATHON 2026 and applicable laws. It is original work of the undersigned/team and has not been copied, wholly or in part, from any other source. The submitted work does not infringe upon any existing intellectual property rights. I/We further declare that the information provided in the submission is true and correct to the best of my/our knowledge.

Mayank Himmatsingka

Name **Designation** **Signature** **Date**